Lecture 8: Rare Disasters and Asset Prices

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Intro

Flight Plan

- Part I: Rietz (1988) as the baseline disaster mechanism (Rietz 1988)
- Part II: Barro (2006) and the empirical discipline step (Barro 2006)
- Part III: Wachter (2013) and time-varying disaster risk (Wachter 2013)
- Part IV: strengths, criticism, and open limits

Why Disaster Models?

- Habit models generate countercyclical prices of risk (Campbell and Cochrane 1999)
- Disaster models target macro left-tail risk directly: you are afraid of large crashes
- Rare severe contractions can move premia a lot
- Core object: how disasters enter the SDF

Part I: Rietz (1988)

What Rietz Tries to Explain

- Standard calibrations miss the observed equity premium (Mehra and Prescott 1985)
- Add a third state to the Mehra and Prescott economy: consumption will fall by a lot
- Keep the representative-agent CRRA setup, with a single tree
- Ask whether tails can reconcile moments (Rietz 1988)
- Everything is the same as Mehra and Prescott (1985) except for the disaster state

Disaster State in Consumption Growth

- y_t is the stochastic dividend from the tree
- Its growth rate $x_t \in \{\lambda_1, \lambda_2, \lambda_3\}$ follows a 3-state Markov chain

$$\lambda_1 = 1 + \mu + \delta, \quad \lambda_2 = 1 + \mu - \delta, \quad \lambda_3 = \psi(1 + \mu)$$

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- The transition matrix is given by:

$$\Phi = \begin{bmatrix} \phi & 1 - \eta - \phi & \eta \\ 1 - \phi - \eta & \phi & \eta \\ 0.5 & 0.5 & 0 \end{bmatrix}$$

- We assume $\lambda_1 > \lambda_2 > \lambda_3$
- η is the probability of a crash

SDF and Tail Risk

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} = \beta \left(\frac{y_{t+1}}{y_t} \right)^{-\gamma}$$

- Rietz noted that the Mehra and Prescott economy was "too symmetric"
- Things are never *that* bad
- Disaster states raise marginal utility sharply
- Payoffs that crash in disasters must pay higher premia

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- Rietz noted that the Mehra and Prescott economy was "too symmetric"
- Things are never *that* bad
- Disaster states raise marginal utility sharply
- Payoffs that crash in disasters must pay higher premia
- You can solve the model exactly like Mehra and Prescott, but now with a 3-state Markov chain

Calibrating The Model

- How to calibrate ψ and η? This is key
- How large should a recession be to be called a “disaster”?
- How frequently can disaster occur?
- By definition this should be super hard to calibrate!

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- How large should a recession be to be called a "disaster"?
- How frequently can disaster occur?
- By definition this should be super hard to calibrate!
- His strategy: given $(\eta, \psi, \gamma, \beta)$, estimate (μ, δ, ϕ) with GMM

Super Large Crash ($\psi = 0.5$)

Example 1: $\lambda_3 = (1 + \mu)/2$.[a] (Output falls to one-half of its normal expected value during a crash.)

Risk aversion and time preference parameter ranges that yield risk premia between 5 and 7 percent with risk-free returns under 3 percent for valid crash probabilities

Crash probability (η)	Risk aversion parameter (α) range	Time preference parameter (β) range
0.0001	—	—
0.0002	8.85–9.00	0.991–0.999
0.0003	8.20–8.50	0.989–0.999
0.0004	7.70–8.10	0.989–0.999
0.0005	7.30–7.80	0.980–0.999
0.0006	7.00–7.60	0.980–0.999
0.0007	6.80–7.35	0.980–0.999
0.0008	6.65–7.20	0.980–0.999
0.0009	6.50–7.05	0.970–0.999
0.0010	6.35–6.85	0.970–0.999
0.0020	5.50–6.00	0.970–0.999
0.0030	5.00–5.45	0.960–0.999
0.0040	4.70–5.15	0.960–0.980

[a] Here λ_3 is the gross growth rate in output during a crash year and $(1 + \mu)$ is the average gross growth rate during 'normal' years.

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Parameter configurations that give risk-free returns and risk premia very near the economy's sample values

Crash probability (η)	Risk aversion parameter (α)	Time preference parameter (β)	Corresponding risk-free return (annual %)	Corresponding risk premium (annual %)
0.0008	7.05	0.997	0.77	6.36
0.0008	7.00	0.999	0.83	6.18
0.0009	6.90	0.994	0.87	6.38
0.0009	6.90	0.995	0.77	6.38
0.0009	6.85	0.997	0.83	6.19
0.0009	6.85	0.998	0.73	6.19
0.0010	6.75	0.993	0.88	6.34
0.0010	6.75	0.994	0.78	6.33
0.0010	6.70	0.996	0.84	6.15
0.0010	6.70	0.997	0.74	6.14
0.0010	6.65	0.999	0.79	5.96
0.0020	5.75	0.989	0.83	5.92
0.0020	5.75	0.990	0.73	5.92
0.0030	5.30	0.980	0.89	6.15

[a] Here λ_3 is the gross growth rate in output during a crash year and $(1+\mu)$ is the average gross growth rate during 'normal' years.

Moderate Crash ($\psi = 0.75$)

Example 2: $\lambda_3 = 0.75$.[a] (Output falls to three-fourths of its previous value during a crash.)

Risk aversion and time preference parameter ranges that yield risk premia between 5 and 7 percent with risk-free returns under 3 percent for valid crash probabilities

Crash probability (η)	Risk aversion parameter (α) range	Time preference parameter (β) range
0.0008	9.75–10.0	0.992–0.999
0.0009	9.40–10.0	0.989–0.999
0.0010	9.10–9.80	0.980–0.999
0.0110	8.80–9.60	0.980–0.999
0.0120	8.55–9.40	0.980–0.999
0.0130	8.30–9.20	0.980–0.999
0.0140	8.15–9.05	0.980–0.999

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0.010	9.80	0.999	0.74	6.95
0.011	9.50	0.999	0.82	6.80
0.012	9.25	0.999	0.84	6.69
0.013	9.15	0.995	0.84	6.86
0.013	9.10	0.997	0.81	6.74
0.013	9.05	0.999	0.78	6.63
0.014	9.00	0.994	0.82	6.83
0.014	8.95	0.996	0.80	6.72
0.014	8.90	0.998	0.77	6.60
0.014	8.85	0.999	0.84	6.49

[a] Here λ_3 is the gross growth rate in output during a crash year.

Questions?

Part II: Barro (2006)

From Rietz (1988) to Barro (2006): What Is New?

- Rietz (1988): proof-of-concept rare catastrophe calibration
- Barro (2006): identifies disaster frequency/severity from long-run cross-country macro data
- Models disaster size as a random variable (distribution), not one fixed drop
- Shows estimated disaster process quantitatively matches equity premium + low r_f
- It is a standard Lucas-tree economy with a few twists: disasters and potential defaults on the risk-free bond
- The state space for consumption growth is now continuous, not discrete

Primitives

- Representative agent with CRRA utility $u(C) = \frac{C^{1-\theta}-1}{1-\theta}$
- Dividend process A_t governed by

$$\log A_{t+1} = \log A_t + \gamma + \sigma u_{t+1} + \nu_{t+1} \log(1 - b_{t+1})$$

where $u_t \sim \mathcal{N}(0,1)$ and $\nu_{t+1} \sim$ Bernoulli$(1 - e^{-p})$, everything i.i.d.

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- Let P_{1t} be the price of the *one-period* dividend claim. Market clearing $C_t = A_t$ implies:

$$P_{1t} = e^{-\rho} \cdot (A_t)^{\theta} \cdot \mathbb{E}_t \left[A_{t+1}^{1-\theta}\right]$$

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- The price-dividend ratio is:

$$\frac{P_{1t}}{A_t} = e^{-\rho} \cdot \mathbb{E}_t \left[\left(\frac{A_{t+1}}{A_t} \right)^{1-\theta} \right]$$

International Evidence on Disasters

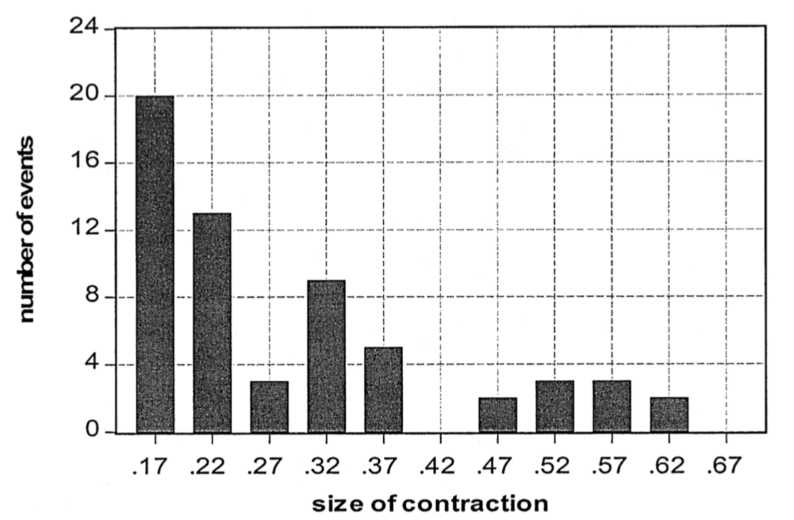
Declines of 15 Percent or More in Real Per Capita GDP

Part A: Twenty OECD Countries in Maddison [2003]

Event	Country	Years	% fall in real per capita GDP
World War I	Austria	1913–1919	35
	Belgium	1916–1918	30
	Denmark	1914–1918	16
	Finland	1913–1918	35
	France	1916–1918	31
	Germany	1913–1919	29
	Netherlands	1913–1918	17
	Sweden	1913–1918	18
Great Depression	Australia	1928–1931	20
	Austria	1929–1933	23
	Canada	1929–1933	33
	France	1929–1932	16
	Germany	1928–1932	18
	Netherlands	1929–1934	16
	New Zealand	1929–1932	18
	United States	1929–1933	31
Spanish Civil War	Portugal	1934–1936	15
	Spain	1935–1938	31
World War II	Austria	1944–1945	58
	Belgium	1939–1943	24
	Denmark	1939–1941	24
	France	1939–1944	49
	Germany	1944–1946	64
	Greece	1939–1945	64
	Italy	1940–1945	45
	Japan	1943–1945	52
	Netherlands	1939–1945	52
	Norway	1939–1944	20
Aftermaths of wars	Canada	1917–1921	30
	Italy	1918–1921	25
	United Kingdom	1918–1921	19
	United Kingdom	1943–1947	15
	United States	1944–1947	28

- He collected international evidence about episodes of 15%+ drops in real per capita GDP
- Both developed in emerging countries (see the paper)

Contraction Size



Fixed Income Gets Screwed Too

STOCK AND BILL RETURNS DURING ECONOMIC CRISES

Event	Real stock return (% per year)	Real bill return (% per year)
World War I		
Austria, 1914–1918	—	−4.1
Denmark, 1914–1918	—	−6.9
France, 1914–1918	−5.7	−9.3
Germany, 1914–1918	−26.4	−15.6
Netherlands, 1914–1918	—	−5.2
Sweden, 1914–1918	−15.9*	−13.1
Great Depression		
Australia, 1928–1930	−3.6	8.2
Austria, 1929–1932	−17.3*	7.1
Canada, 1929–1932	−23.1*	7.1
Chile, 1929–1931	−22.3*	—
France, 1929–1931	−20.5	1.4
Germany, 1928–1931	−14.8	9.3
Netherlands, 1929–1933	−14.2*	5.7
New Zealand, 1929–1931	−5.6*	11.9
United States, 1929–1932	−16.5	9.3
Spanish Civil War		
Portugal, 1934–1936	13.4*	3.8
World War II		
Denmark, 1939–1945	−3.7*	−2.2
France, 1943–1945	−29.3	−22.1
Italy, 1943–1945	−33.9	−52.6
Japan, 1939–1945	−2.3	−8.7
Norway, 1939–1945	1.7*	−4.5
Post-WWII Depressions		
Argentina, 1998–2001	−3.6	9.0
Chile, 1981–1982	−37.0*	14.0
Indonesia, 1997–1998	−44.5	9.6
Philippines, 1982–1984	−24.3	−5.0
Thailand, 1996–1997**	−48.9	6.0
Venezuela, 1976–1984	−8.6*	—

- Depression times: stocks go bad, bonds do well
- Wars: everything goes bad
- Are bonds that risk-free?

Introducing Bond Defaults

- In a disaster, the risk-free bond may default with probability q
- Upon a default, the holder looses a fraction d

$$R^b_{1,t} = \begin{cases} R^f_{1,t} & \text{with no disasters, with probability } e^{-p} \\ R^f_{1,t} & \text{with disasters, but no default, with probability } (1-e^{-p})(1-q) \\ (1-d)R^f_{1,t} & \text{with disasters and default, with probability } (1-e^{-p})q \end{cases}$$

- In the paper: $d = b$ as a simplifying assumption
- The distribution of b and d are independent of the other shocks and independent over time

Solving The Model

- Define the gross return on equity as $R^e_{1,t+1} = \frac{A_{t+1}}{P_{1,t}}$
- Exercise: show that the expected equity return is such that

$$\log\left[\mathbb{E}_t\left(R^e_{t+1}\right)\right] = \rho + \theta\gamma - \frac{1}{2}\theta^2\sigma^2 + \theta\sigma^2 - p\left[\mathbb{E}\left((1-b)^{1-\theta}\right) - 1 + \mathbb{E}b\right].$$

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- The return conditional on no disasters is (please show it too):

$$\log\left(\mathbb{E}_t\left[R^e_{t+1} \mid \nu_{t+1} = 0\right]\right) = \rho + \theta\gamma - \frac{1}{2}\theta^2\sigma^2 + \theta\sigma^2 - p\left[\mathbb{E}\left((1-b)^{1-\theta}\right) - 1\right]$$

Solving The Model

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- The risk-free rate is (show this at home too):

$$\log\left[\mathbb{E}_t\left(R^b_{t+1}\right)\right] = \rho + \theta\gamma - \frac{1}{2}\theta^2\sigma^2 - p\left[(1-q)\,\mathbb{E}\left((1-b)^{-\theta}\right) + q\,\mathbb{E}\left((1-b)^{1-\theta}\right) + q\,\mathbb{E}b - 1\right]$$

The Equity Premium

The risk premium is define as $\log(\mathbb{E}[R^e_{1t}]) - \log(\mathbb{E}[R^b_{1t}])$

$$\log(\mathbb{E}[R^e_{1t}]) - \log(\mathbb{E}[R^b_{1t}]) = \theta\sigma^2 + p(1-q)\left[\mathbb{E}((1-b)^{-\theta}) - \mathbb{E}((1-b)^{1-\theta}) - \mathbb{E}b\right]$$

- The first term is the usual one: you're compensated for systematic risk
- The second term increases in p and decreases in q. Why?
- The premium is increasing in $\mathbb{E}[b]$ as well. Why?

What if there is leverage?

- Leveraged firms are riskier: equity holder are *residual* claimants
- Payments to equity holders are even more procyclical
- Leverage makes equity look like a Call Option on the firm's assets

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- Consider a firm with a debt-to-equity ratio of λ
- Firm's debt pays the risk-free rate, but may "default" if there is a disaster
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- Consider a firm with a debt-to-equity ratio of λ
- Firm's debt pays the risk-free rate, but may "default" if there is a disaster
- Upon a disaster, a fraction d of the assets will disappear with probability q
- Barro (2006) shows that the levered equity premium is given by

$$\text{levered premium} = (1+\lambda)\left\{\theta\sigma^2 + p(1-q)\left[\mathbb{E}\left((1-b)^{-\theta}\right) - \mathbb{E}\left((1-b)^{1-\theta}\right) - \mathbb{E}b\right]\right\}$$

- Using American data, he calibrates it at $\lambda = 0.5$

Main Result

Calibrated Model for Rates of Return

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
			Parameters				
	No disasters	Baseline	Low θ	High p	Low q	Low γ	Low ρ
θ (coeff. of relative risk aversion)	4	**4**	**3**	4	4	4	4
σ (s.d. of growth rate, no disasters)	0.02	**0.02**	0.02	0.02	0.02	0.02	0.02
ρ (rate of time preference)	0.03	**0.03**	0.03	0.03	0.03	0.03	**0.02**
γ (growth rate, deterministic part)	0.025	**0.025**	0.025	0.025	0.025	**0.020**	0.025
p (disaster probability)	**0**	**0.017**	0.017	**0.025**	0.017	0.017	0.017
q (bill default probability in disaster)	**0**	**0.4**	0.4	0.4	**0.3**	0.4	0.4
			Variables				
Expected equity rate	0.128	0.071	0.076	0.044	0.071	0.051	0.061
Expected bill rate	0.127	0.035	0.061	−0.007	0.029	0.015	0.025
Equity premium	0.0016	0.036	0.016	0.052	0.042	0.036	0.036
Expected equity rate, conditional	0.128	0.076	0.081	0.052	0.076	0.056	0.066
Face bill rate	0.127	0.037	0.063	−0.004	0.031	0.017	0.027
Equity premium, conditional	0.0016	0.039	0.019	0.056	0.045	0.039	0.039
Price-earnings ratio	9.7	19.6	17.8	37.0	19.6	27.8	24.4
Expected growth rate	0.025	0.020	0.020	0.018	0.020	0.015	0.020
Expected growth rate, conditional	0.025	0.025	0.025	0.025	0.025	0.020	0.025
			Levered results (debt-equity ratio is λ = 0.5)				
Expected equity rate	0.129	0.089	0.084	0.071	0.092	0.069	0.079
Equity premium	0.0024	0.054	0.024	0.078	0.063	0.059	0.054
Expected equity rate, conditional	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Not Only Fun and Games

The variance of the growth rate of A_t is given in (24). In the baseline model in Table V, column (2), the price-earnings ratio is constant. Therefore, the standard deviation of stock returns equals the standard deviation of the growth rate of A_t. When $p = 0.017$, this standard deviation equals 0.046 or 0.053 when disaster sizes b have the distribution given by Figure I, panel A or B, respectively. These values apply to samples with representative numbers of disasters, such as the long periods considered in the upper part of Table III. However, the average standard deviation of stock returns over these periods was 0.23 (Table IV), way above the value predicted by the model. Similarly, the tranquil period 1954–2004 displayed in the lower part of Table III should corre-

A Conjecture

relative to the volatility of returns. My conjecture is that the introduction of variations in p_t would generate a closer match between observed and theoretical Sharpe ratios. That is, the

Questions?

Part III: Wachter (2013)

Wachter's Core Extension

- Keep disaster-size mechanism from Barro
- Add time-varying disaster probability (Wachter 2013)
- Disaster risk becomes a persistent state variable
- Target: premia, valuation ratios, and volatility dynamics

Time-Varying Disaster Intensity

$$\lambda_{t+1} = (1 - \phi_\lambda)\bar{\lambda} + \phi_\lambda \lambda_t + \sigma_\lambda \sqrt{\lambda_t}\, \varepsilon^{\lambda}_{t+1}$$

- λ_t: conditional disaster intensity
- Persistence creates long-lived disaster scares
- Shocks to λ_t move discount rates

Consumption Growth with Stochastic Intensity

$$\Delta \log C_{t+1} = g + \sigma\varepsilon_{t+1} + \nu_{t+1} \log(1 - b_{t+1}), \qquad \mathbb{P}_t(\nu_{t+1} = 1) = \lambda_t$$

- Same jump-size logic as Barro
- Disaster arrival risk varies over time
- Conditional tail risk drives conditional premia

Valuation Implication

$$\frac{P_t}{D_t} = f(\lambda_t), \qquad \frac{\partial f(\lambda_t)}{\partial \lambda_t} < 0$$

- High disaster intensity lowers price-dividend ratios
- Expected returns rise in high-λ_t states
- Volatility and predictability move with tail-risk state

How Wachter Took It to Data

- Calibrate persistence and volatility of λ_t
- Match premium and volatility moments jointly
- Compare valuation-ratio dynamics with data
- Evaluate return-predictability implications

Quantitative Fit: What Improves?

- Better conditional dynamics than constant-p models
- Richer movements in valuation ratios
- Stronger time variation in expected returns
- Fit remains sensitive to tail-process assumptions

TODO Figure: Wachter Disaster Intensity State

- TODO: insert disaster-intensity figure from Wachter (2013)
- Intended exhibit: high-λ_t episodes and valuation compression

TODO Table: Wachter Quantitative Fit

- TODO: insert benchmark moments table from Wachter (2013)
- Intended exhibit: data vs model on premia, volatility, predictability

Questions?

Part IV: Pros, Cons, and Criticism

Pros of the Disaster-Risk Agenda

- Coherent macro-finance tail-risk mechanism
- Joint intuition for high premium and low average r_f
- Clear mapping from disasters to SDF movements
- Stronger empirical discipline from Rietz to Barro/Wachter

Criticism and Limits

- Rare-event identification is fragile in finite samples
- Disaster definitions vary across countries and periods
- Quantitative fit is sensitive to tail-size assumptions
- Not all cross-asset moments are matched simultaneously

Closing

Takeaways

- Rietz: tails can solve premium puzzles in principle
- Barro: key innovation is data-disciplined disaster frequencies and sizes
- Wachter: time-varying disaster probability adds conditional dynamics
- Main debate: quantitative fit vs tail-risk identification limits

AI-Generated Summary

- Disaster models price macro tail risk directly
- Barro (2006) moved the agenda from calibrated thought experiment to measured tails
- Wachter (2013) added a disaster-risk state variable
- Strengths and criticism both hinge on tail measurement

Readings and References i

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